DoseRAD2026
Real-time Dose Calculation Challenge

Endorsed by ESTRO, PTCOG, EFOMP, DGMP, NVKF, BZKF, The Hellenic Association of Medical Physicists (HAMP), Svensk Förening för Radiofysik, Svenska Sjukhusfysiker Förbundet

Sponsored by: RaySearch Laboratories, RAD formation

Hosted by: MICCAI2026

MICCAI2026
29TH INTERNATIONAL CONFERENCE ON MEDICAL IMAGE COMPUTING & COMPUTER ASSISTED INTERVENTION
Sept. 27- Oct. 1, 2026
Strasbourg Convention Center, FRANCE

DoseRAD2026 Grand Challenge: Summary and Closing Statements



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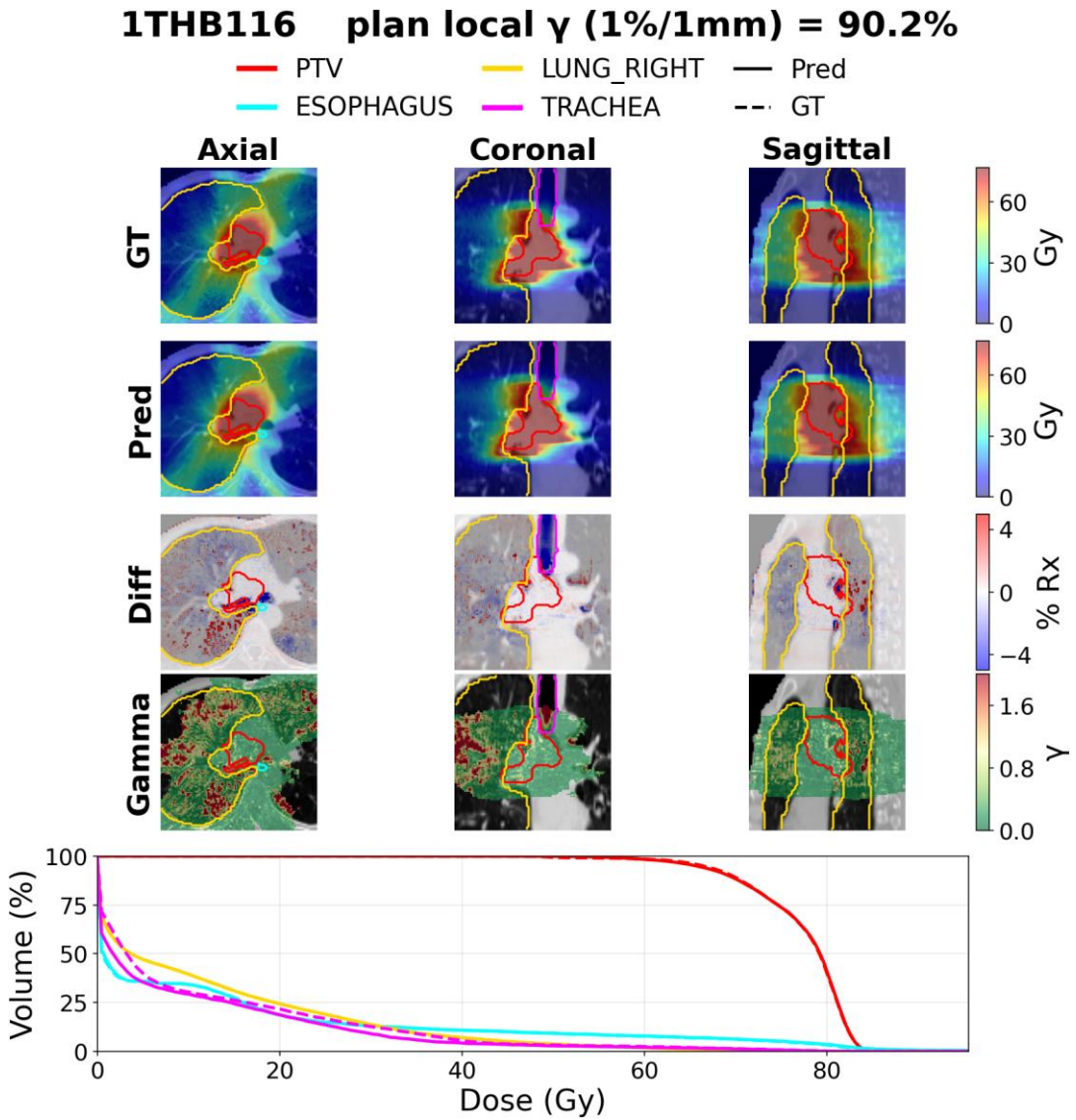
MICCAI 2026 Satellite Event,
Strasbourg, France
October 1st, 2026

Summary

What have we achieved?

- **Photon CT champion**
  - **Accuracy:** plan local γ 98.4%, beam MAE 0.68%
  - **Speed:** 23.0 s per plan (181 segments)
  - **Median case:** AUMC23, plan γ 98.9%
  - **Worst case:** 1THB116, plan γ 90.2%

Metric	Mean	Median	Worst
Plan local γ 1%/1mm (%)	98.4	98.9	90.2
Beam MAE (% of beam max)	0.68	0.67	0.93
DVH score	0.24	0.17	1.27
Time per segment (ms)	127		
Patients / segments		37 / 6697	

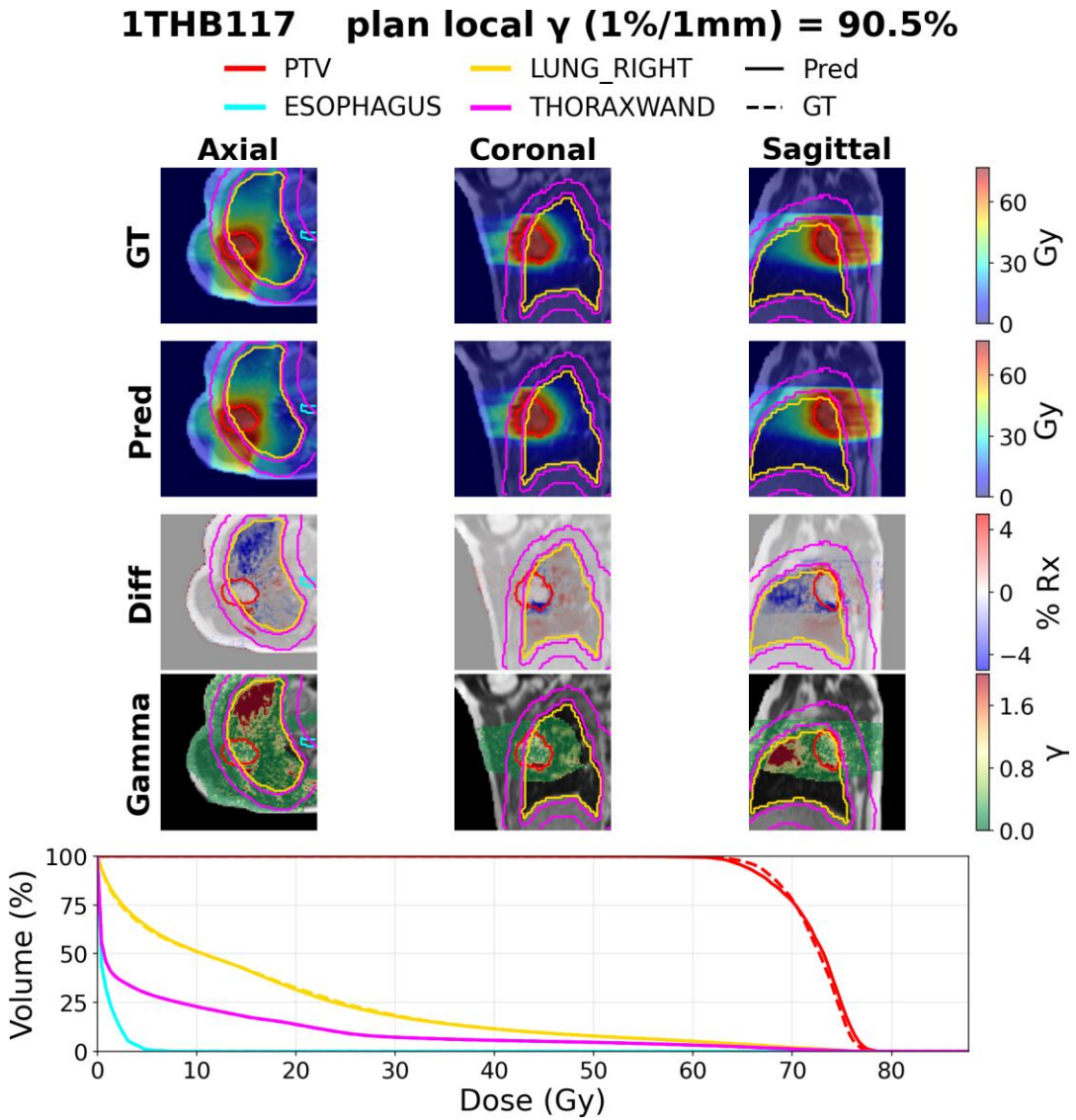


Summary

What have we achieved?

- **Photon MRI champion**
  - **Accuracy:** plan local γ 88.5%, beam MAE 1.37%
  - **Speed:** 36.2 s per plan (181 segments)
  - **Median case:** 1THB117, plan γ 90.4%
  - **Worst case:** AUMC32, plan γ 62.0%

Metric	Mean	Median	Worst
Plan local γ 1%/1mm (%)	88.5	90.4	62.0
Beam MAE (% of beam max)	1.37	1.22	3.39
DVH score	2.92	0.89	20.65
Time per segment (ms)	200		
Patients / segments		37 / 6697	

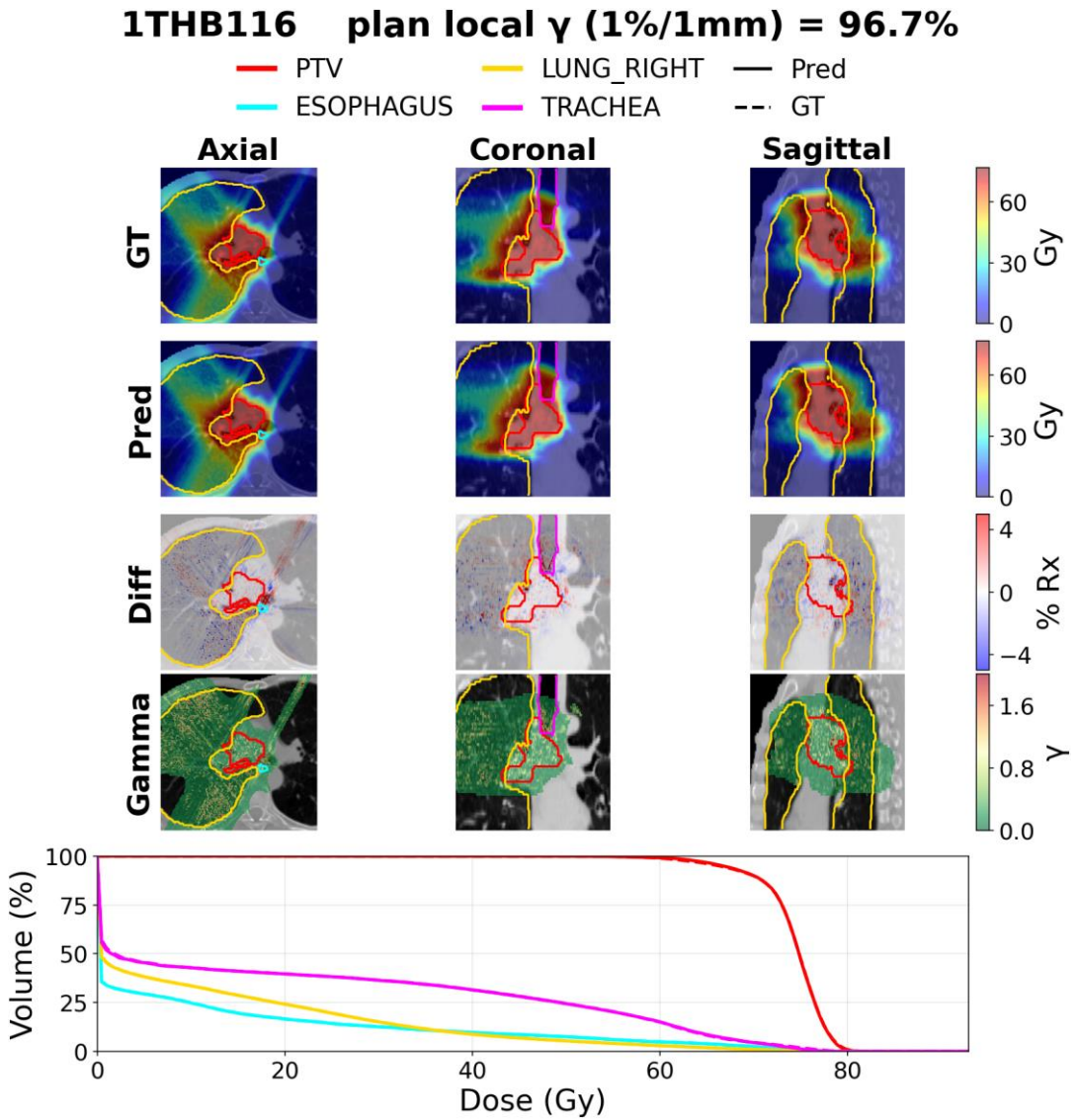


Summary

What have we achieved?

- **Proton CT champion**
  - **Accuracy:** plan local γ 99.4%, beam MAE 0.48%
  - **Speed:** 88.0 s per plan (500 beamlets)
  - **Median case:** 1THB104, plan γ 99.5%
  - **Worst case:** 1THB116, plan γ 96.7%

Metric	Mean	Median	Worst
Plan local γ 1%/1mm (%)	99.4	99.5	96.7
Beam MAE (% of beam max)	0.48	0.48	0.75
DVH score	0.22	0.18	0.65
Time per beamlet (ms)	176		
Patients / beamlets		37 / 27140	

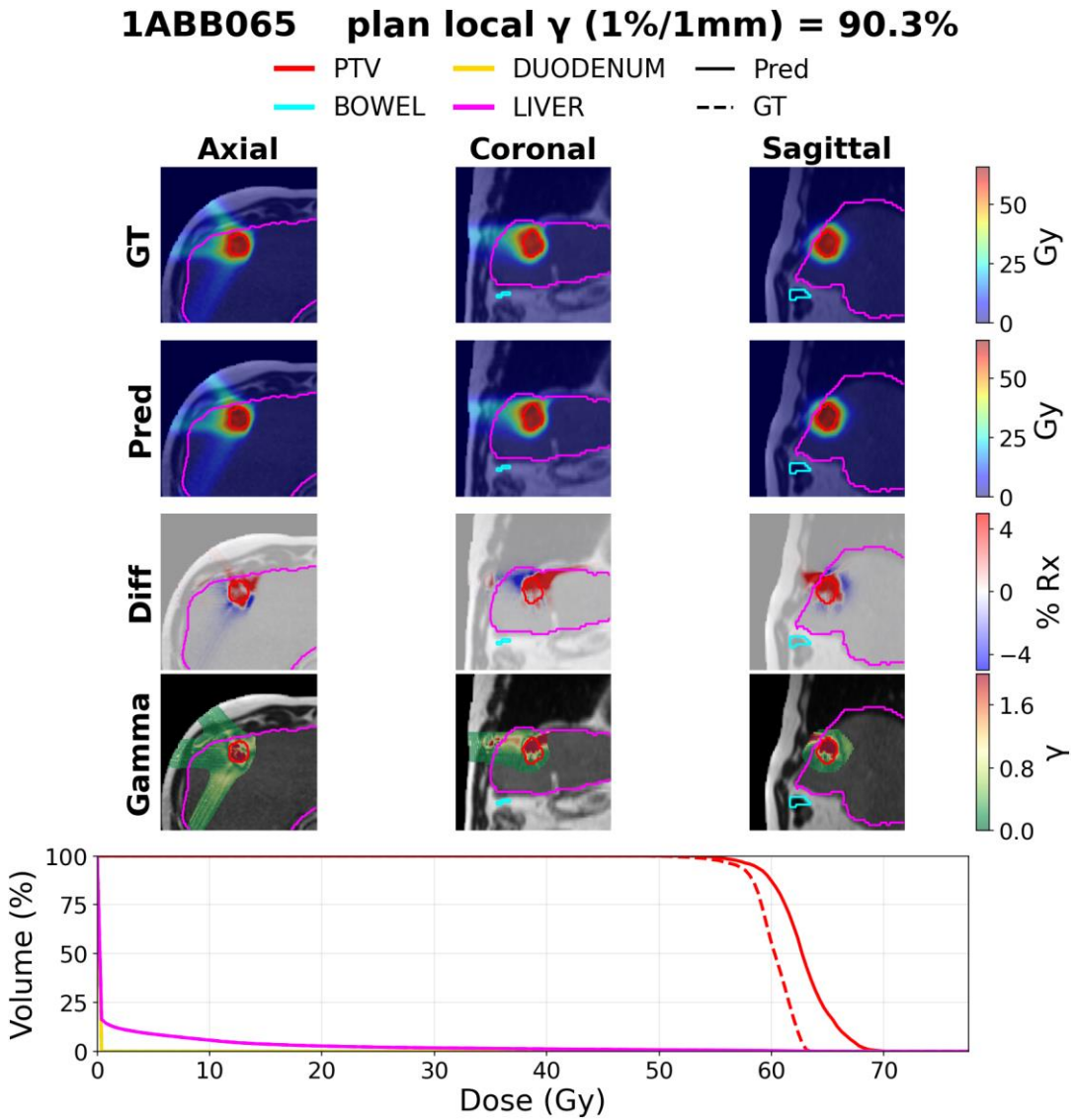


Summary

What have we achieved?

- **Proton MRI champion**
  - **Accuracy:** plan local γ 85.6%, beam MAE 1.69%
  - **Speed:** 95.7 s per plan (500 beamlets)
  - **Median case:** 1ABB065, plan γ 90.3%
  - **Worst case:** AUMC32, plan γ 47.2%

Metric	Mean	Median	Worst
Plan local γ 1%/1mm (%)	85.6	90.3	47.2
Beam MAE (% of beam max)	1.69	1.43	5.38
DVH score	3.81	1.35	33.81
Time per beamlet (ms)	191		
Patients / beamlets		37 / 27140	



Summary

Overview of all tasks

- **CT is solved; MRI is the bottleneck**
  - MRI fails in the lung, most at the external centre (AUMC)
  - Same hardest patients across tasks: 1THB116 (CT), AUMC32 (MRI)

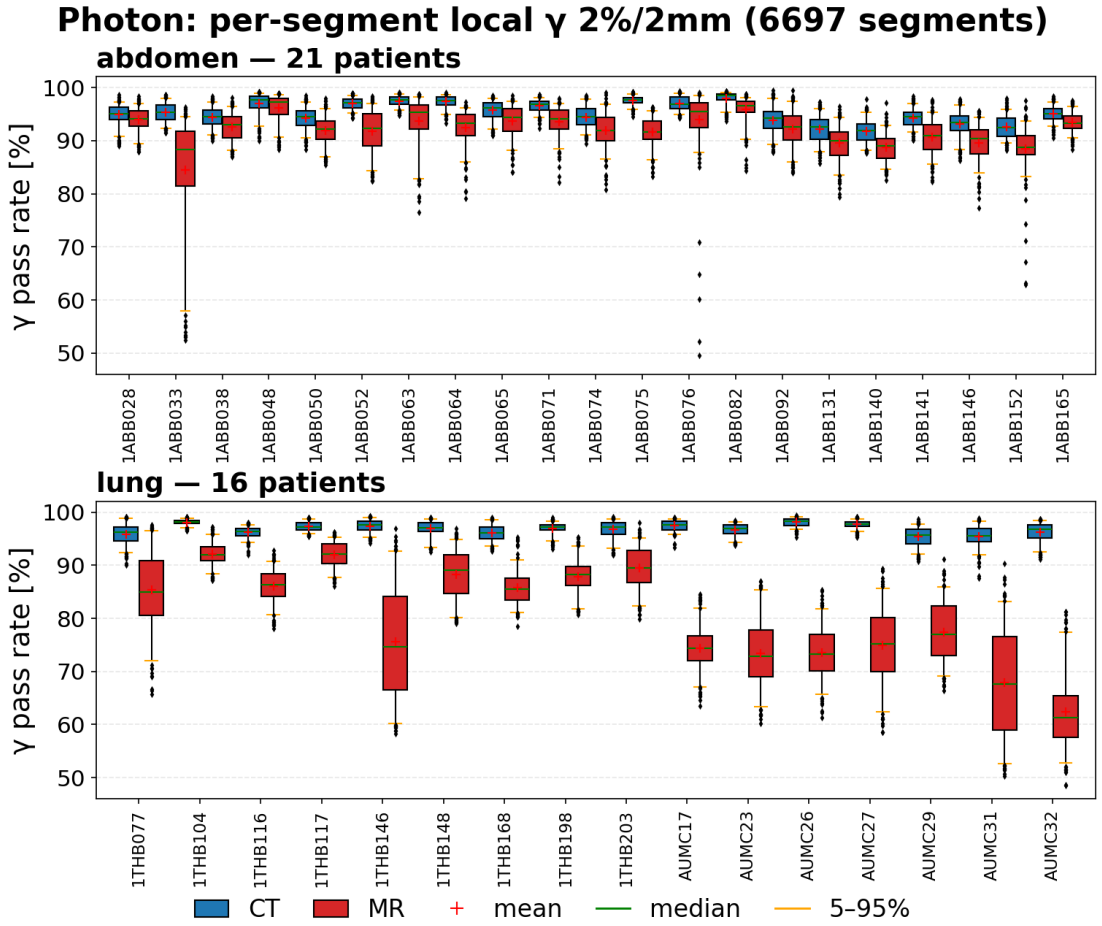
Task	Plan local γ 1%/1mm (%): mean (worst)				Beam MAE mean (% of beam max)	DVH score	Time per plan (s)	
	All (n=37)	LMU abdomen (n=21)	LMU lung (n=9)	AUMC lung (n=7)			GC	Local*
Photon CT	98.4 (90.2)	98.9 (97.6)	96.9 (90.2)	98.9 (98.3)	0.68	0.24	23.0 (181)	16.5 (181)
Photon MRI	88.5 (62.0)	95.3 (87.3)	84.8 (75.1)	**73.0 (62.0)**	1.37	2.92	36.2 (181)	19.9 (181)
Proton CT	99.4 (96.7)	99.6 (98.6)	99.0 (96.7)	99.3 (98.9)	0.48	0.22	88.0 (500)	51.0 (500)
Proton MRI	85.6 (47.2)	92.0 (75.6)	84.5 (67.2)	**67.9 (47.2)**	1.69	3.81	95.7 (500)	56.8 (500)

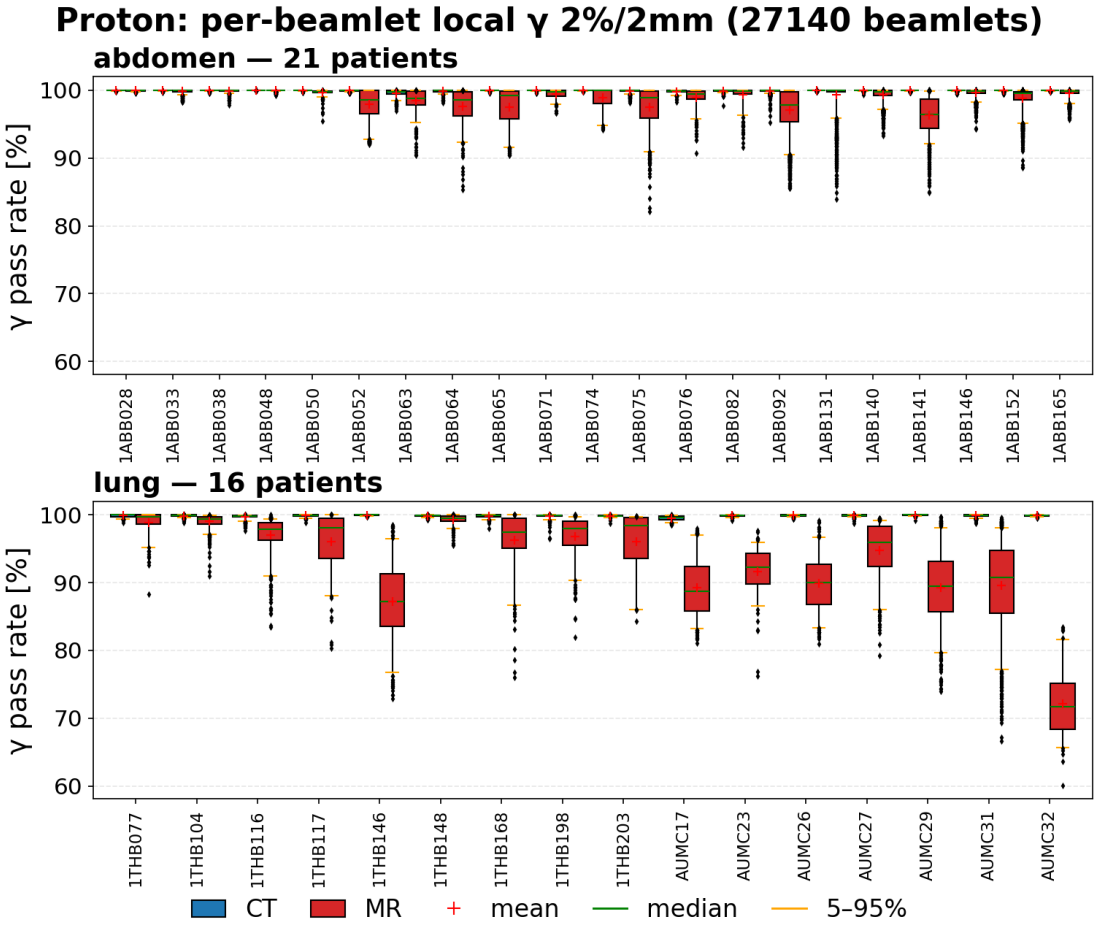
GC: Champion models' Grand Challenge leaderboard time.
Local*: Champion models on a local NVIDIA RTX A6000 GPU (48 GB) with 4 CPUs; plan accuracy unchanged.

Summary

Overview of all tasks

- MRI dose calculation in lung is still challenging

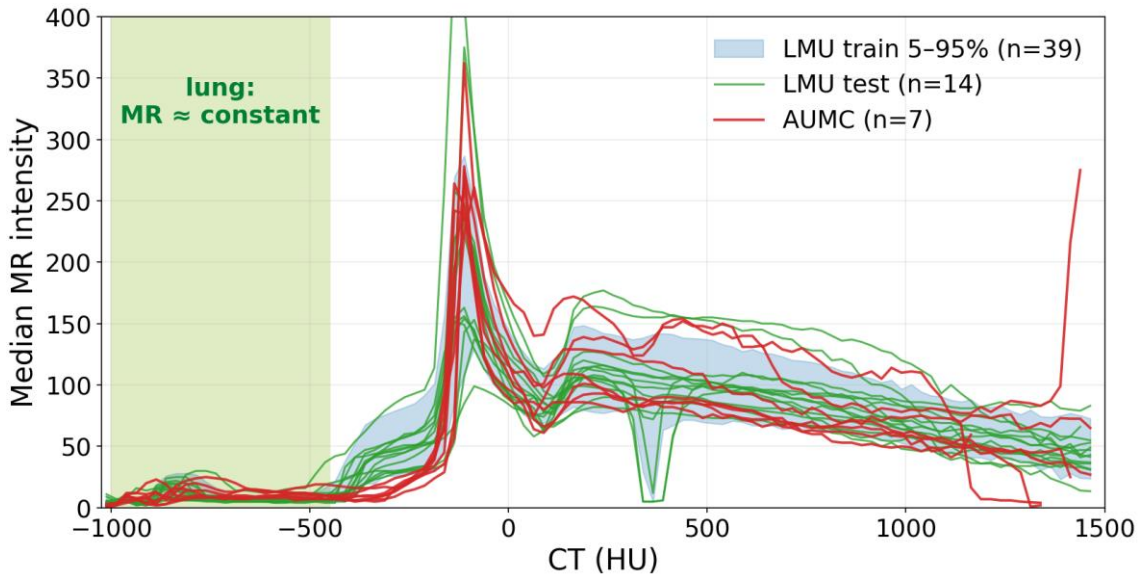


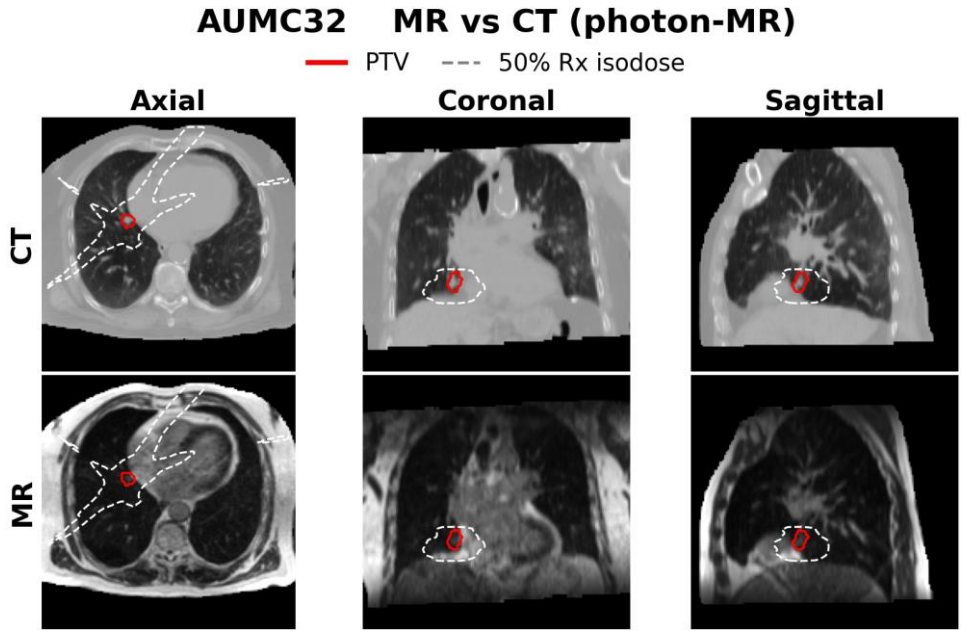


Summary

Why does MRI fail in the lung?

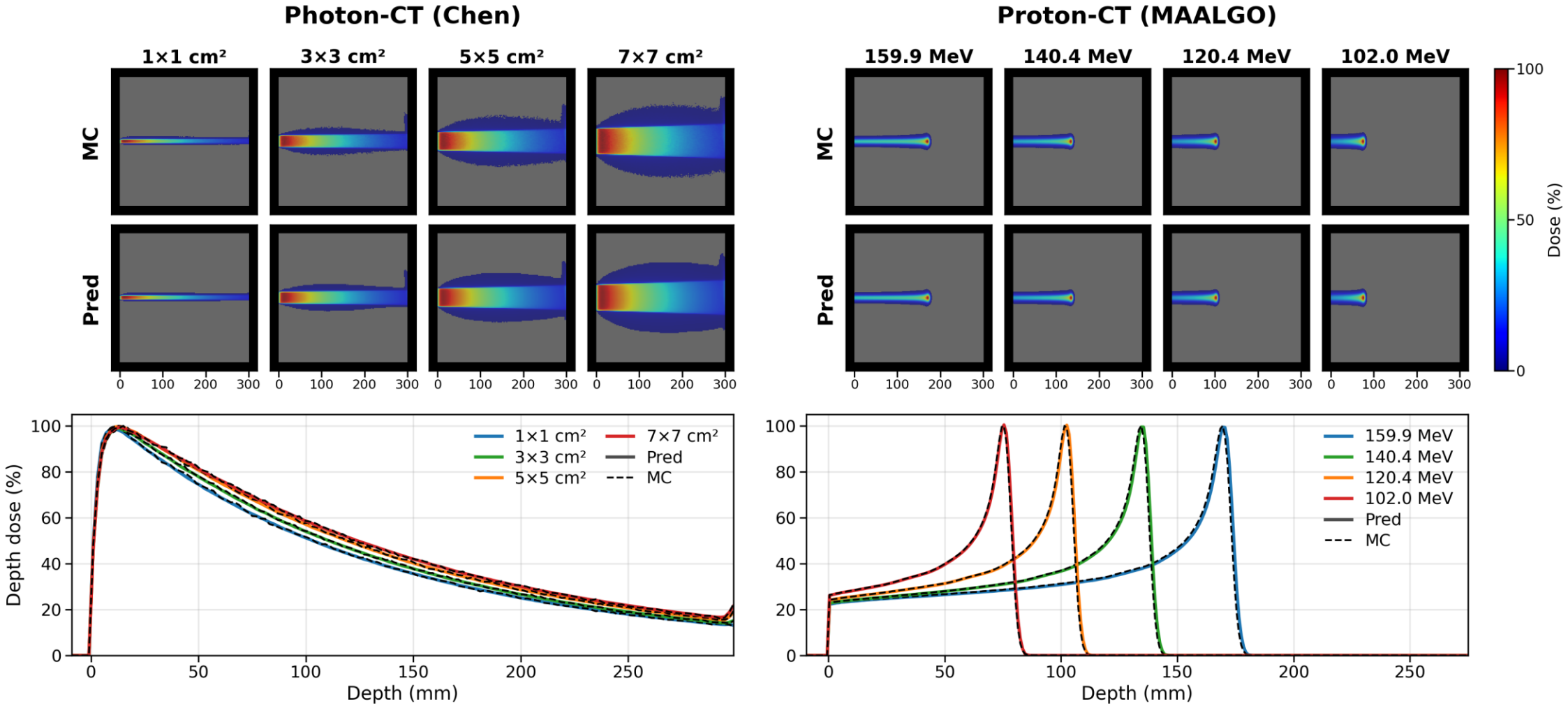
- Lung dose depends on lung density: on CT it spans −1000 to −450 HU
- MR is flat over that whole range (left): the model can only guess the density
- Scanner shift? AUMC MR (red) lies inside the LMU training band (blue)
- Worst case AUMC32 (right): small tumor, clear on CT, barely visible on MR





Challenge goals

Independent phantom test



Challenge goals
Summary

- **Conclusion**
  - **CT tasks:** fast and accurate
  - **MRI tasks:** abdomen close to CT, the lung challenge is still open
  - **DoseRAD test dataset** together with SynthRAD test data will be released in March 2030

- **Outlook**
  - GPU-MC vs Champion DL models
  - Json -> segment/beamlet dose & plan dose
  - MC modelling of clinical linacs
  - Images from more sites and different vendors
  - CBCT dose tasks

DoseRAD2026
Thank you!

Guillaume Landry



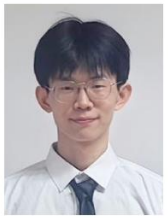
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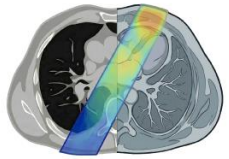
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